Lipids

UNIT XVIII



Learning Outcomes

At the end of each topic, the students should be able to:

- ❑ Define lipids and classify lipids
- ❑ Define, identify, and classify fatty acids
- ❑ Describe fatty acids' physical properties.
- ❑ Define and classify triacylglycerols as simple or mixed
- ❑ Explain chemical reactions that triacylglycerols undergo

Lipids

- **Lipid**: An organic compound found in living organisms that is insoluble (or only sparingly soluble) in water but soluble in non-polar organic solvents
- Unlike other biomolecules, lipids do not have a common structural feature that serves as the basis for defining such compounds
- Classification is based on two methods
  - Biochemical function
  - Structure

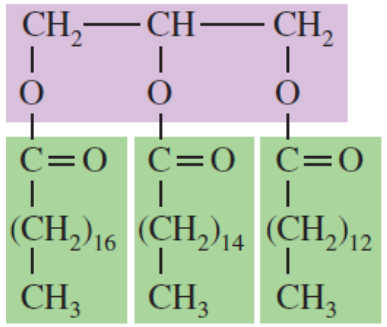
Classification according to Biological Functions

1. **Energy-storage lipids** – triacylglycerol
    - those containing hydrocarbon side chains serve as energy stores
2. **Membrane lipids** – phospholipid, sphingoglycolipids, and cholesterol
    - Structural components of cell membranes that serves as *protective barrier*
    - Protective functions in bacteria, plants, insects, & vertebrates, serving as a part of the outer coating between the body of the organism and the environment.
3. **Emulsification lipids** – bile acids
4. **Messenger lipids** – steroid hormones and eicosanoids
    - Many intra- and intercellular signaling events involve lipid molecules
5. **Protective coating** - waxes

TYPES OF LIPIDS according to Structures

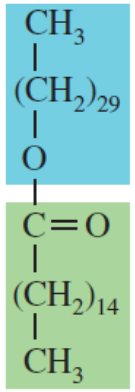
1. **Simple lipids.** These consist of long chain fatty acids with an alcohol by an ester linkage. They include the triglycerides (triacylglycerols) and the waxes.
2. **Complex lipids** which contain additional groupings such as phosphoric acid, sugars, nitrogenous bases or proteins. Included in this group are the phospholipids, glycolipids and lipoproteins.
3. **Derived lipids** (Steroids, Prostaglandins, Thromboxanes, and leukotrienes & Terpenes). Although they do not often contain fatty acids the steroids are frequently classed as lipids on account of their occurrence in natural fats and their solubility characteristics. They include cholesterol and the sex and adrenocortical hormones.

SIMPLE



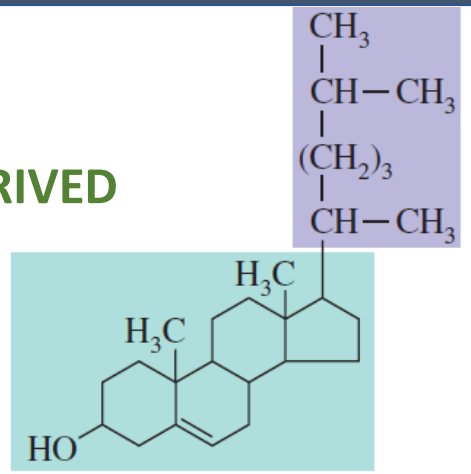
A fat

SIMPLE



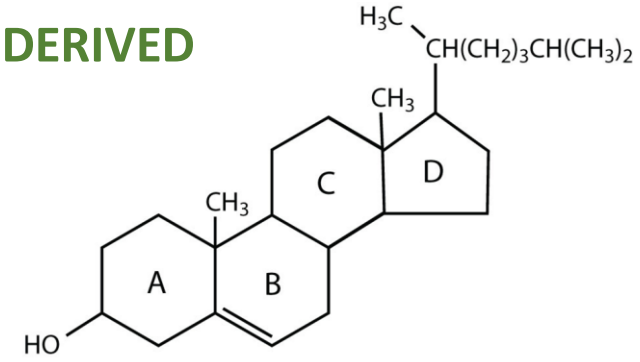
A biological wax

DERIVED



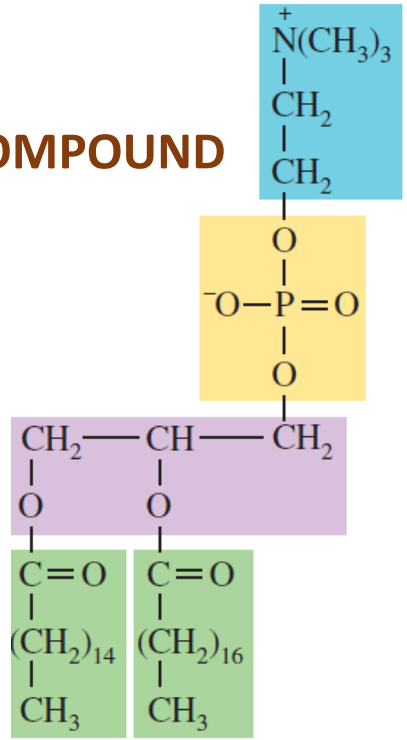
A steroid

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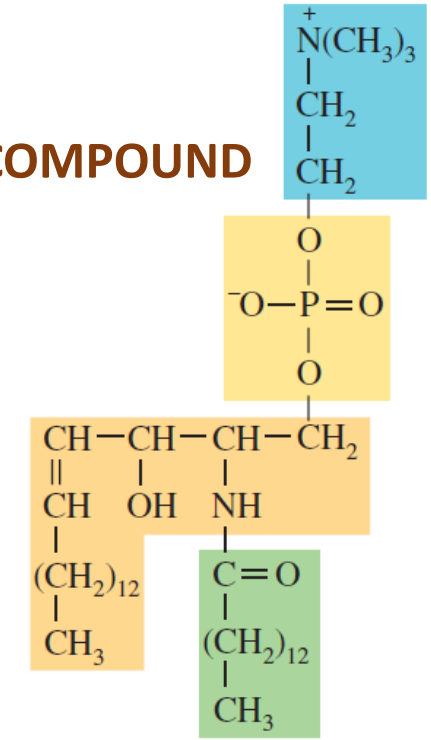
(b) Cholesterol

COMPOUND



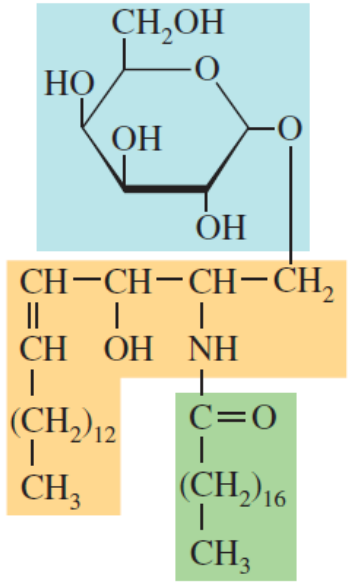
glycerophospholipid

COMPOUND



A sphingophospholipid

COMPOUND



A sphingoglycolipid

Figure 19.1 - Structural Formula of Types of Lipids

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Exercise

A lipid is an organic compound found in living organisms that is:

a.soluble in water and insoluble in nonpolar organic solvents.

b.soluble in water and soluble in nonpolar organic solvents.

c.insoluble in water and insoluble in nonpolar organic solvents.

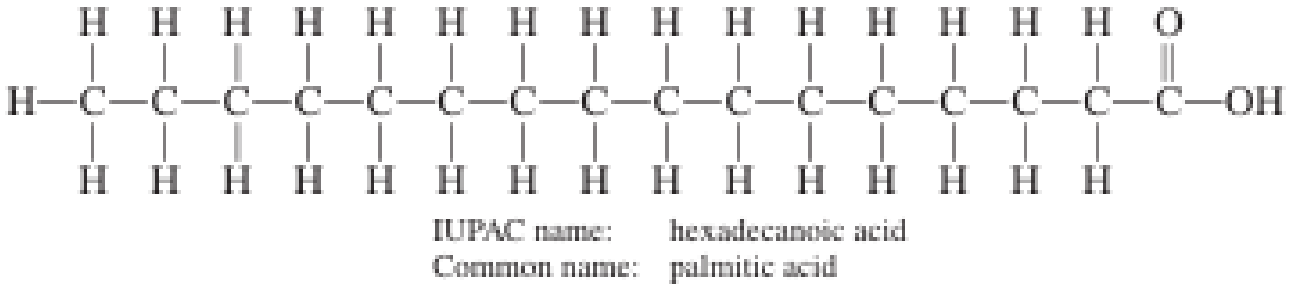
d.insoluble in water and soluble in nonpolar organic solvents.

Fatty Acids

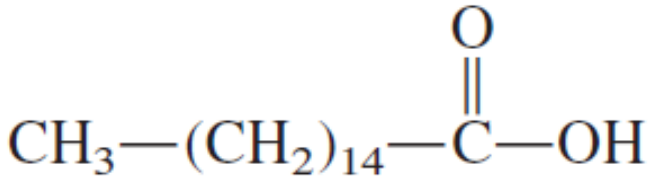
- **Fatty acids**: a molecule composed of a **long hydrocarbon chain** and a **carboxyl group (-COOH)** at one end, acting as a building block for fats and other lipids.
- Fatty acids are *amphipathic* compounds because the carboxyl group is hydrophilic, and the hydrocarbon tail is hydrophobic.
- It is used by the body for energy, is a component of cell membranes, and can be classified as **saturated** (single bonds only) or **unsaturated** (one or more double bonds)

Saturated Fatty Acids

- Fatty acids with a carbon chain in which all **C–C bonds are single bonds**
- Numbering starts from the end of –COOH group
- Consider the structural notations for palmitic acid:



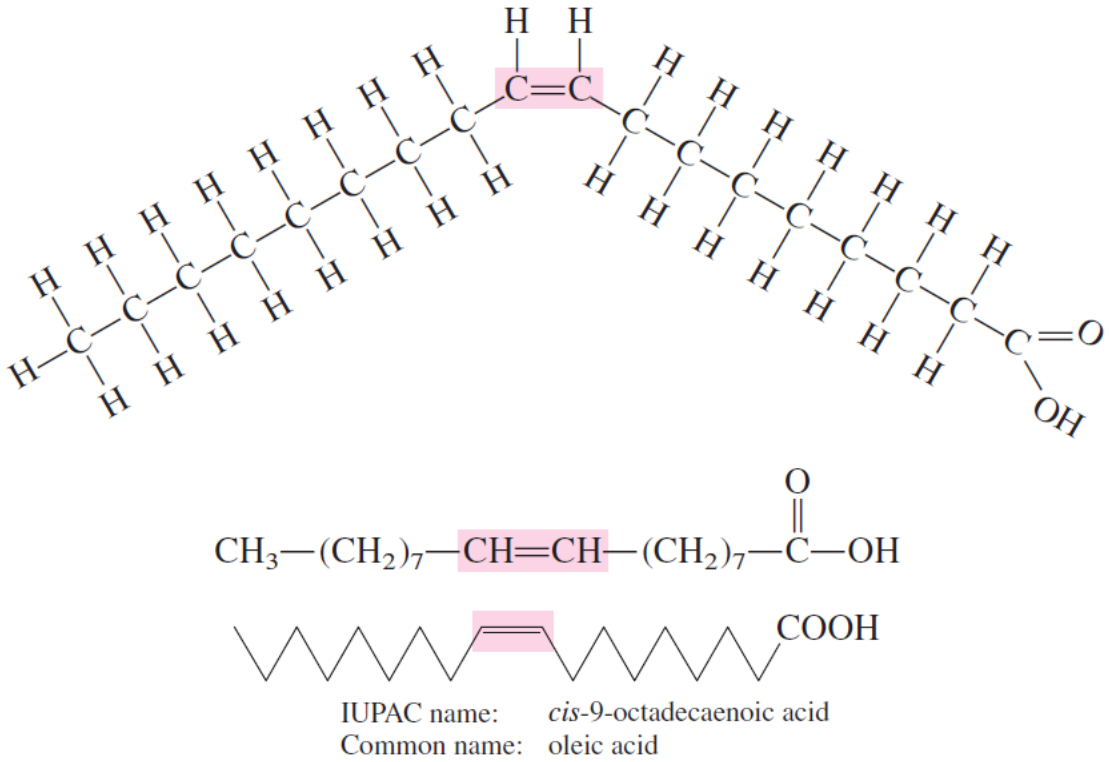
IUPAC name: hexadecanoic acid
Common name: palmitic acid





Unsaturated Fatty Acids

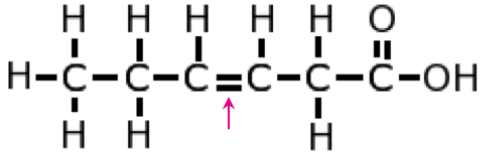
- **Unsaturated fatty acid**: Fatty acid with a carbon chain in which one carbon–carbon double bond is present
- There are different ways of depicting the structure



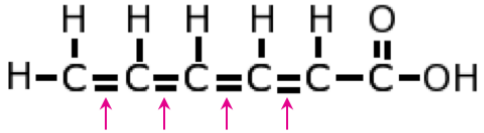
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2. **Unsaturated Fatty Acid** - is a fatty acid with a carbon chain has at least one carbon–carbon double bond

    1. Monounsaturated Fatty Acid - contains one carbon–carbon double bond
    2. Polyunsaturated Fatty Acid - contains two or more carbon–carbon double bonds
    3. Omega-3 Fatty Acid - its endmost double bond is three carbon atoms away from its methyl end
    4. Omega-6 Fatty Acid - its endmost double bond is six carbon atoms away from its methyl end.

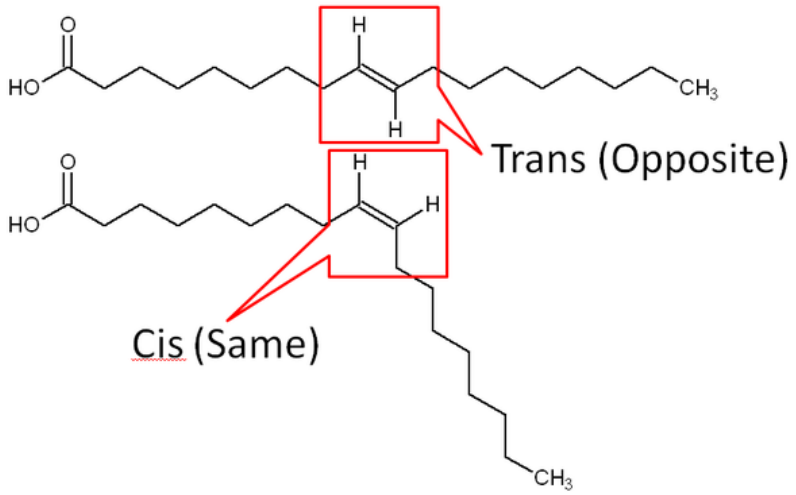


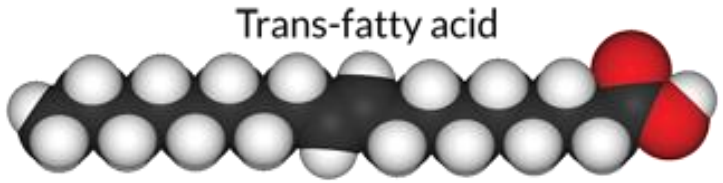
Monounsaturated fatty acid

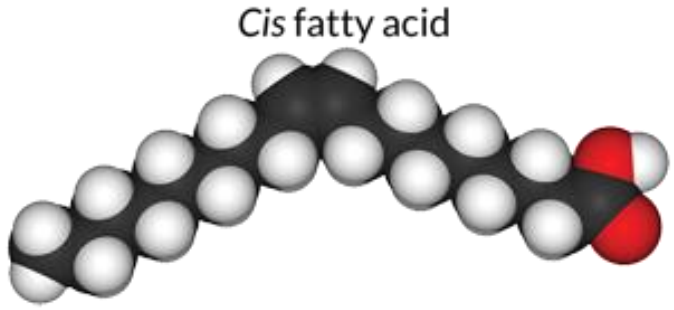


Polyunsaturated fatty acid

Cis and Trans Fatty Acids







Properties	Cis Fatty acids	Trans Fatty acids
Health implications	Good	Detrimental; lowers good cholesterol and increases the level of bad cholesterol in the body.
Natural occurrence	Yes	Very less, hence produced artificially by partial hydrogenation of polyunsaturated fatty acids.
Orientation at double bonded carbon atoms	Hydrogen atoms at double bonded carbon atoms on same side	Hydrogen atoms at double bonded carbon atoms on opposite side
Recommended consumption	Can consume as per requirment	No more than 1% of total calories per day
Commonly found in	Natural fatty acids	Processed food, fast foods, butter and milky products.
Melting point	Lower	Higher

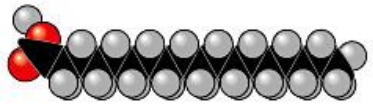
https://courses.lumenlearning.com/suny-nutrition/chapter/2-32-fatty-acids/

Fats and Oils

FAT is a triacylglycerol mixture that is a solid or semi-liquid at room temperature (25-degree Celsius). Generally, fats are obtained from animal sources.

Fats are composed of triacylglycerols in which saturated fatty acids predominate, although some unsaturated fatty acids are present.

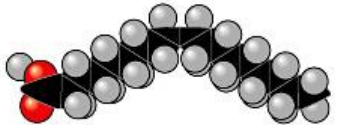




OIL is a triacylglycerol mixture that is a liquid at room temperature (25 degrees Celsius). Generally, oils are obtained from plant sources.

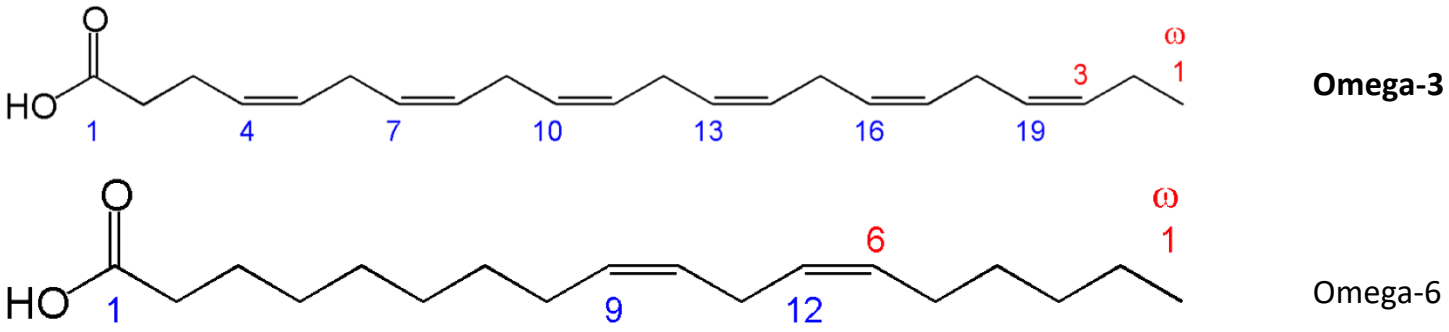
Oils contain triacylglycerols with larger amounts of mono- and polyunsaturated fatty acids than those in fats.





Essential Fatty Acids

- Fatty acids that must be obtained from dietary sources because they cannot be synthesized within the body
- Two most important essential fatty acids are:
  - Linoleic acid (18:2) - **Omega 6**
  - Linolenic acid (18:3) - **Omega 3**
    - Needed for proper membrane structure
    - Serve as starting materials for the production of several nutritionally important longer-chain omega-6 and omega-3 fatty acids



Omega-3

Omega-6

Essential and Non-essential Fatty Acids

1. **Nonessential Fatty acids -** Fatty acids that can be synthesized in the body

2. **Essential Fatty acids -** Fatty acids that must be obtained from the diet; these are precursors of eicosanoids
    1. Omega-6 fatty acid
    2. Omega-3 fatty acid

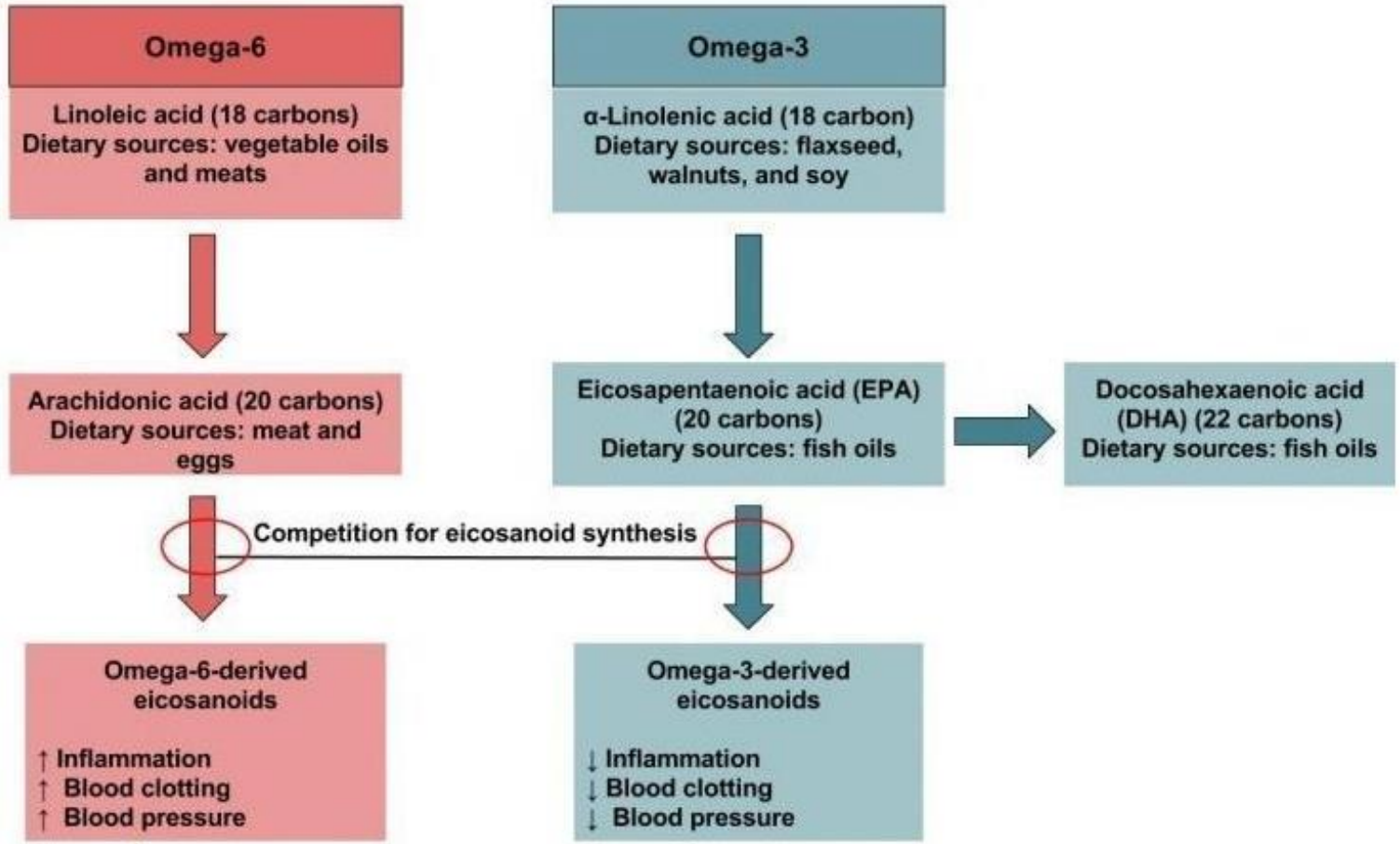


Figure 5.5.1: Essential Fatty Acids. *Image by Allison Calabrese* / CC BY 4.0.

https://med.libretexts.org/Under_Construction/Purgatory/Book%3A_Human_Nutrition_(University_of_Hawaii)_1st_Ed/05%3A_Lipids/5.05%3A_Nonessential_and_Essential_Fatty_Acids

EXERCISE

A fatty acid is a naturally occurring _____ acid that nearly always contains an _____ number of carbon atoms with a carbon chain that is _____.

a. monocarboxylic; even; branched

b. dicarboxylic; odd; unbranched

c. monocarboxylic; even; unbranched

d. dicarboxylic; even; branched

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EXERCISE

Which of the following best describe the physical properties of fatty acids?

a. Melting points of fatty acids are influenced by the carbon chain length.

b. Melting points of fatty acids are influenced by their degree of unsaturation.

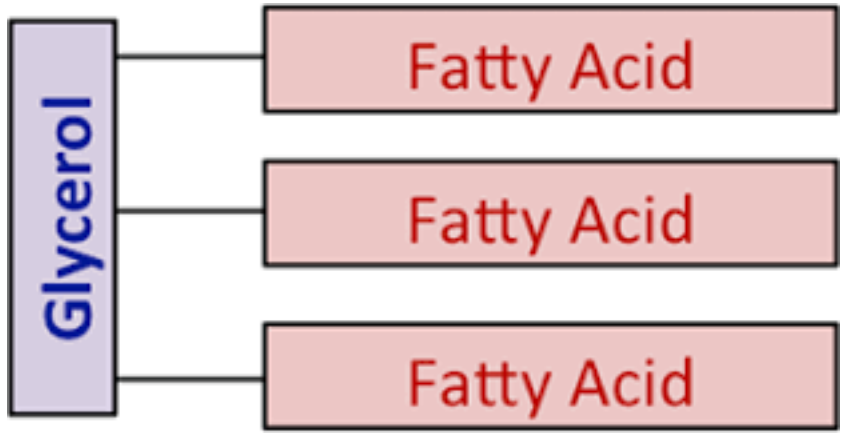
c. All physical properties of fatty acids are influenced by the length and degree of unsaturation of their carbon chain.

d. All the above.

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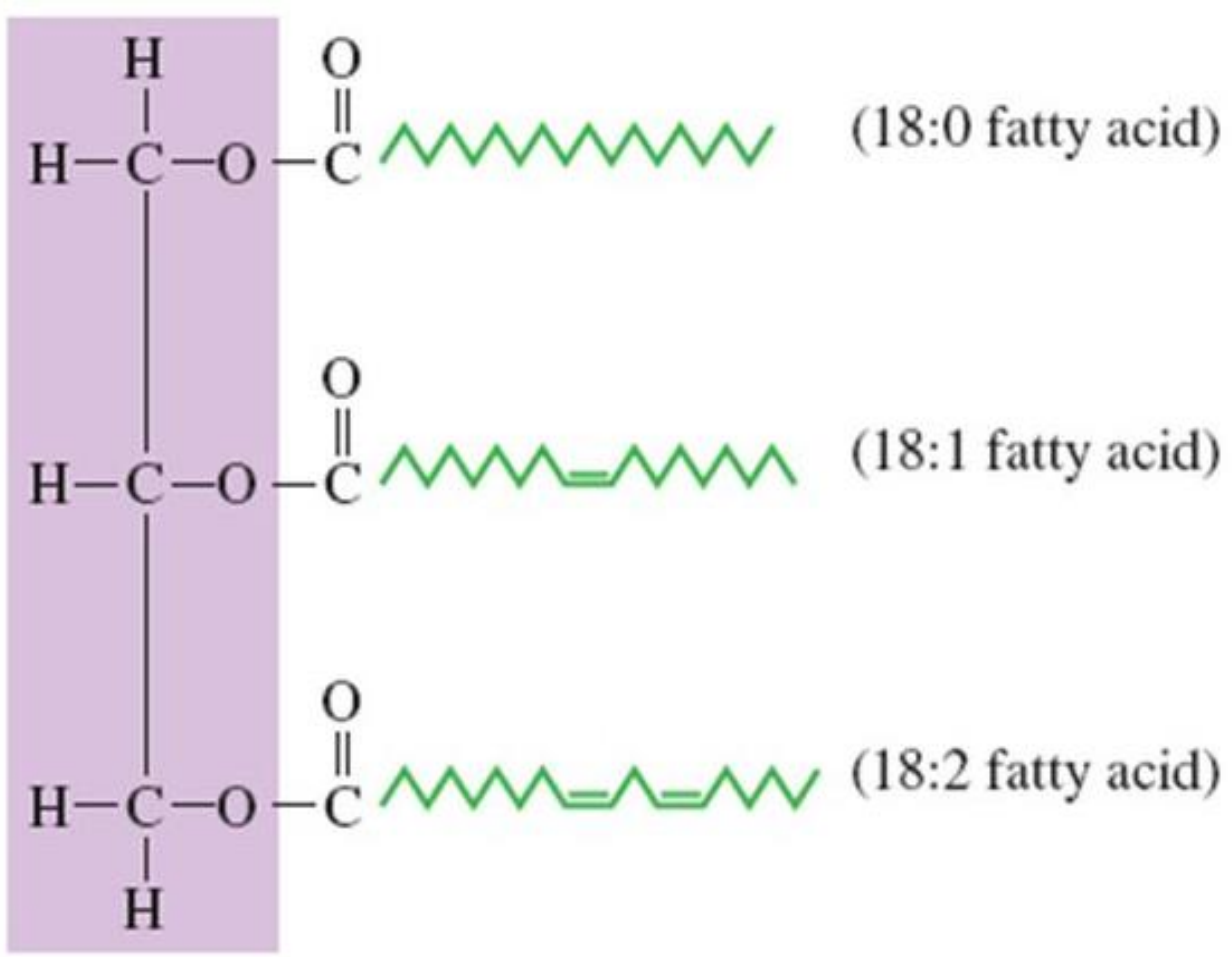
Triacylglycerols (triglycerides)

- Lipid that contains a glycerol molecule and three fatty acids bonded together by ester bond
- They function within the body as energy-storage materials.
- Triacylglycerols are concentrated primarily in special cells (adipocytes) that are nearly filled with the material.
- Adipose tissue containing these cells is found in various parts of the body: under the skin, in the abdominal cavity, in the mammary glands, and around various organs. Functions as insulator and shock absorber.
- The fat content of normal humans allows them to survive starvation for 2 or 3 months



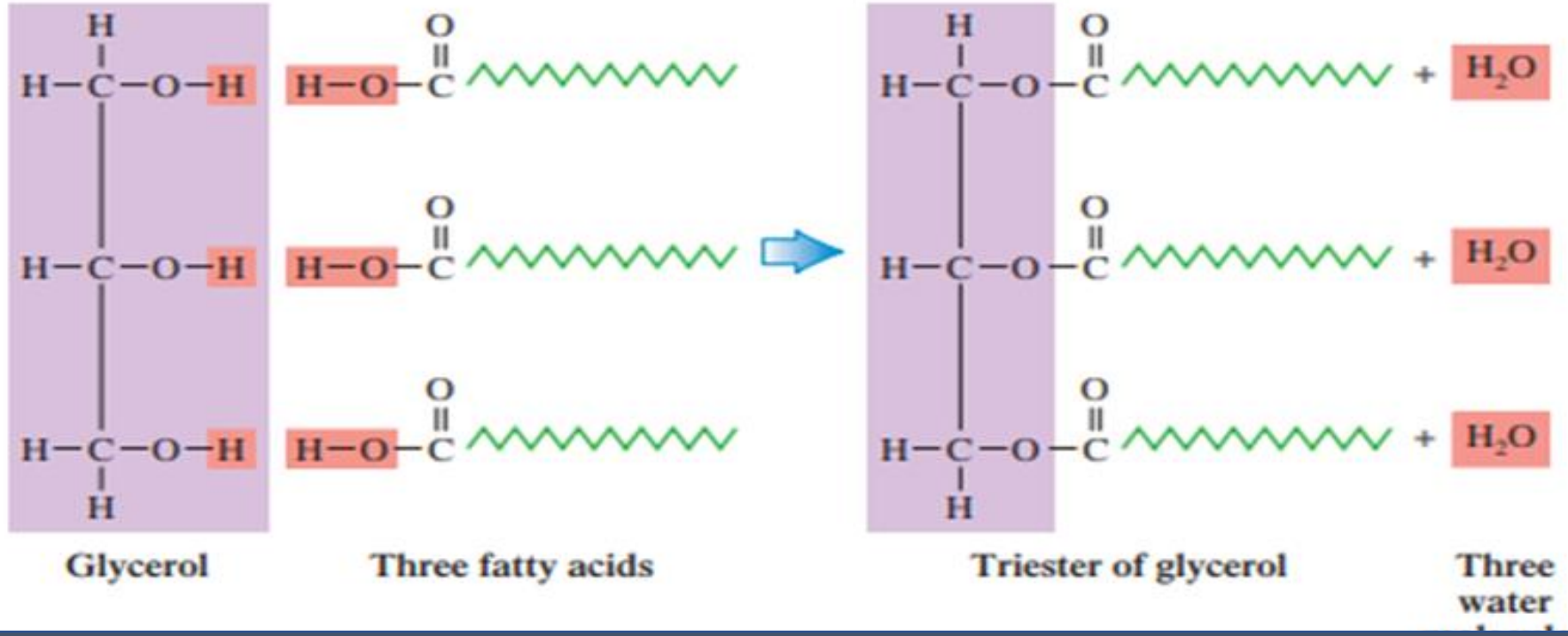
Types of Triacylglycerols

- **Simple triacylglycerol** is a triester formed from the esterification of glycerol with three *identical* fatty acid molecules.
- **Mixed triacylglycerol** is a triester formed from the esterification of glycerol with *more than one kind* of fatty acid molecule.



Esterification

In the **esterification reaction** a single molecule of glycerol reacts with three fatty acid molecules each of three hydroxyl groups present is esterified to produce a triacylglycerol.



Exercise

Which of the following statements is true about the classification of fatty acids?

a. They are classified as saturated, monounsaturated, or polyunsaturated fatty acids.

b. They are classified as saturated or branched unsaturated fatty acids.

c. They are classified as saturated or polyunsaturated fatty acids.

d. None of the above.

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- Bettelheim, F., Brown, W., Campbell, M., Farrell, S., Torres, O.J., & Madsen, S. (2020) Introduction to General, Organic and Biochemistry (12th ed.). Cengage.
- Brown, T. L., LeMay, H. E., Bursten, B. E., Murphy, C. J., & Woodward, P. M. (2012). Chemistry: The Central Science (12th ed.). Pearson Prentice Hall.
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